MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – CHAPTER TEST (SET – 2)** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, complete solutions and step marking are provided for both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. The distance of the point $(3, 5)$ from the x-axis is: [1]

(A) 3 **(B)** 5 **(C)** 8 **(D)** $\sqrt{34}$

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 5

Given: Point $P(3, 5)$. Perpendicular foot on the x-axis is $M(3, 0)$.

Formula Used: Distance of (x, y) from x-axis $= |y|$

$$PM = \sqrt{(3-3)^2 + (0-5)^2} = \sqrt{0+25} = |5| = \boxed{\mathbf{5}\text{ units}}$$

Q2. M is a point on the y-axis at a distance of 4 units from the x-axis and it lies below the x-axis. The distance of point M from point $Q(5, 1)$ is: [1]

(A) $\sqrt{2}$ units **(B)** $\sqrt{34}$ units
(C) $\sqrt{50}$ units **(D)** $\sqrt{90}$ units

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) $\sqrt{50}$ **units**

Given: Point M lies on the y-axis ($x = 0$), at 4 units **below** the x-axis $\implies y = -4$, so $M(0, -4)$. Second point is $Q(5, 1)$.

Formula Used: $MQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$$MQ = \sqrt{(5-0)^2 + (1-(-4))^2} = \sqrt{5^2 + (1+4)^2} = \sqrt{25+25} = \boxed{\sqrt{\mathbf{50}}\text{ units}}$$

Q3. The line represented by $2y - x = 4$ intersects the y-axis at: [1]

(A) $(2, 0)$ **(B)** $(0, -4)$ **(C)** $(0, 2)$ **(D)** $(2, 2)$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) $(0, 2)$

Given: Equation of line $2y - x = 4$.

Formula Used: For intersection with the y-axis, put $x = 0$.

Substituting $x = 0$:

$$2y - 0 = 4 \implies 2y = 4 \implies y = 2$$

$$\boxed{\text{Intersection point } = (\mathbf{0}, \mathbf{2})}$$

Q4. The distance of the point $P(-4, 3)$ from the y-axis is: [1]

(A) 4 **(B)** -4 **(C)** 3 **(D)** 5

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 4

Given: Point $P(-4, 3)$. Perpendicular foot on the y-axis is $N(0, 3)$.

Formula Used: Distance of (x, y) from y-axis $= |x|$

$$PN = \sqrt{(0 - (-4))^2 + (3 - 3)^2} = \sqrt{4^2 + 0} = |-4| = \boxed{\mathbf{4} \text{ units}}$$

Q5. The distance of the point $(4, 0)$ from the x-axis is: [1]

(A) 4 units **(B)** 16 units **(C)** 0 units **(D)** $4\sqrt{2}$ units

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) 0 units

Given: Point $(4, 0)$ has y-coordinate $y = 0$, which means it lies directly on the x-axis itself.

Formula Used: Distance of (x, y) from x-axis $= |y|$

$$\text{Distance} = |0| = \boxed{\mathbf{0} \text{ units}}$$

SECTION B: Short Answer Type Questions

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Q6. Find the distance between the points $P(-5, 7)$ and $Q(-1, 3)$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Formula Declaration [1 Mark]

Given: Points $P(-5, 7)$ and $Q(-1, 3)$.

Formula Used: $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Step 2: Substitution and Simplification [2 Marks]

$$PQ = \sqrt{(-1 - (-5))^2 + (3 - 7)^2}$$

$$= \sqrt{(-1 + 5)^2 + (-4)^2} = \sqrt{4^2 + (-4)^2} = \sqrt{16 + 16} = \sqrt{32} = \sqrt{16 \times 2} = \boxed{\mathbf{4\sqrt{2}} \text{ units}}$$

Q7. Find the distance of a point $P(x, y)$ from the origin. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Identifying Coordinates of Origin [1 Mark]

Given: Point $P(x, y)$ and the origin $O(0, 0)$.

Formula Used: $OP = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Step 2: Applying the Distance Formula [2 Marks]

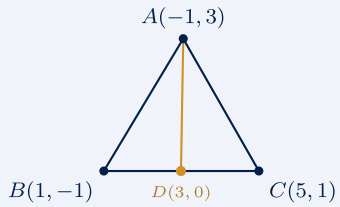
Substituting $(x_1, y_1) = (0, 0)$ and $(x_2, y_2) = (x, y)$:

$$OP = \sqrt{(x - 0)^2 + (y - 0)^2} = \boxed{\sqrt{\mathbf{x^2 + y^2}} \text{ units}}$$

SECTION C: Long Answer & Competency Based Questions (9 Marks)

Q8. The vertices of a triangle are $A(-1, 3)$, $B(1, -1)$ and $C(5, 1)$. Find the length of the median through vertex A. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



Step 1: Finding Midpoint D of Opposite Side BC [2 Marks]

Let AD be the median through vertex $A(-1, 3)$, so D is the midpoint of side BC with $B(1, -1)$ and $C(5, 1)$. By the midpoint formula, the coordinates of D average those of B and C, after which the distance formula yields median length AD.

We know the midpoint and distance formulas:

Formula Used: $D = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$ and $AD = \sqrt{(x_D - x_A)^2 + (y_D - y_A)^2}$

Computing the coordinates of midpoint D:

$$D = \left(\frac{1 + 5}{2}, \frac{-1 + 1}{2}\right) = \left(\frac{6}{2}, \frac{0}{2}\right) = (\mathbf{3}, \mathbf{0})$$

Step 2: Finding Length of Median AD [3 Marks]

Using the distance formula between $A(-1, 3)$ and $D(3, 0)$:

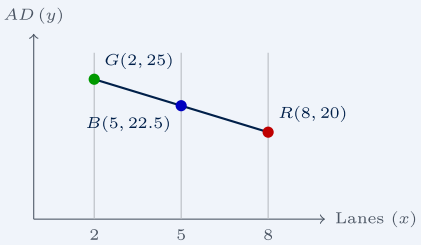
$$AD = \sqrt{(3 - (-1))^2 + (0 - 3)^2} = \sqrt{(3 + 1)^2 + (-3)^2} = \sqrt{4^2 + (-3)^2} = \sqrt{16 + 9} = \sqrt{25} = \boxed{\mathbf{5} \text{ units}}$$

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Q9. Competency Case Study: To conduct Sports Day activities in a rectangular school ground $ABCD$, lines have been drawn with chalk powder at a distance of 1 m each. 100 flower pots have been placed at a distance of 1 m from each other along AD. Niharika runs $\frac{1}{4}$th the distance AD on the 2nd line and posts a green flag. Preet runs $\frac{1}{5}$th the distance AD on the 8th line and posts a red flag. [4]

(i) Write the coordinates of the green flag posted by Niharika.
(ii) Write the coordinates of the red flag posted by Preet.
(iii) Find the distance between both the flags.
(iv) If Rashmi has to post a blue flag exactly halfway between the line segment joining the two flags, where should she post her flag?

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]



(i) Coordinates of Green Flag (G) [1 Mark]

$x = 2$ (2nd line), $y = \frac{1}{4} \times 100 = 25$ m $\implies$ $\boxed{\mathbf{G(2, 25)}}$

(ii) Coordinates of Red Flag (R) [1 Mark]

$x = 8$ (8th line), $y = \frac{1}{5} \times 100 = 20$ m $\implies$ $\boxed{\mathbf{R(8, 20)}}$

(iii) Distance Between Both Flags (GR) [1 Mark]

We know the distance and midpoint formulas:

Formula Used:
$$GR = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$GR = \sqrt{(8-2)^2 + (20-25)^2} = \sqrt{6^2 + (-5)^2} = \sqrt{36 + 25} = \boxed{\sqrt{\mathbf{61}} \text{ m}}$$

(iv) Position of Rashmi's Blue Flag (B) [1 Mark]

Since the blue flag is halfway between $G(2, 25)$ and $R(8, 20)$, using the midpoint formula:

$$B = \left(\frac{2+8}{2}, \frac{25+20}{2} \right) = \left(\frac{10}{2}, \frac{45}{2} \right) = (\mathbf{5},\ \mathbf{22.5})$$

Hence, she should post the blue flag at:

$\boxed{\mathbf{5}\text{th line at a distance of } \mathbf{22.5} \text{ m above } \mathbf{AB}}$

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